

Statistical Learning

Home Assignment 1

Question 2

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Assessing the Usability of Accelerometer Data in Detecting Real-Time Behaviour of a Domestic Animal Species

1. Introduction

Keeping track of what animals do is important for research, animal care, and farm work. Watching them by eye or studying video takes a lot of time and different observers may disagree. Because of this, small wearable sensors like accelerometers are becoming popular. They can record the animal's movements every second and help computers sort those movements into clear behavior categories.

An accelerometer senses how fast something speeds up or slows down in three directions (left-right, forward-back, up-down). This lets us capture both the way the animal holds its body and how it moves (activity).

This study aims to evaluate whether the data extracted from an accelerometer can accurately classify the behavior of a domestic animal species using statistical learning methods. Behavior labels were derived from video annotations according to predefined classes within 5-second windows. To answer this question, we applied Linear Discriminant Analysis (LDA) and k-Nearest Neighbors (KNN) to assess

how well accelerometer data alone can identify behaviors like walking, resting, grazing, or running.

Since the output must be placed into clear behaviour labels, we treat the task as a classification problem. To check whether the accelerometer can be trusted, we let a model predict the behaviours and then compare those predictions with the true labels. If the model reaches a high accuracy on the test set, we can conclude that the accelerometer data predicts the animal's behaviour well.

2. Data and Method

Methods and Data

Data Collection and Description

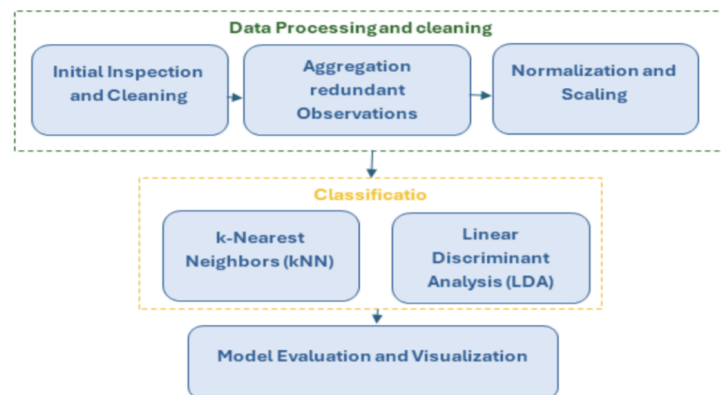
The behaviors in this study were labeled every five seconds from video and a behavior guide. The data, in the form of a file called "df4_5s.csv," consisted of features from the motion sensors and was used to test how well different models can recognize animal behaviors.

Data Preprocessing and Cleaning

Data cleaning took place in R with the help of tidyverse packages, **dplyr**, and **lubridate**. As the first step, the dataset was loaded and visually checked for any structural irregularities. To aid in future temporal analyses, the Timestamp column was converted to POSIXct date time form. Other ID columns such as animal ID or behavior label ID were transformed into categorical

We used grouping by sample, timestamp, and modifiers as identifiers to remove redundant observation data. For each of these groups, the average was calculated over all numerical features. This aggregation helped ensure that each record corresponded to a unique pattern of behavior over time for each member of the sample population. We used z-score normalization to change the distribution of all numerical variables to a

Figure 1. Data Processing and Classification Workflow



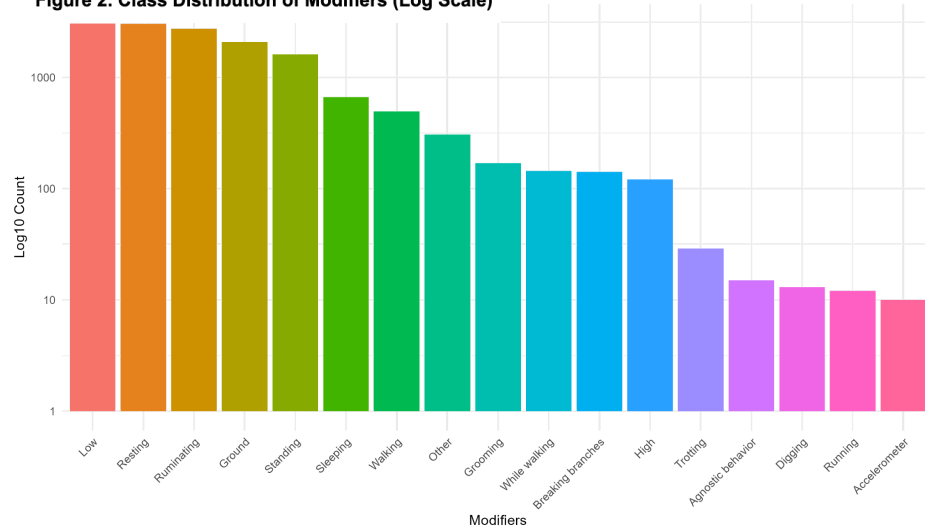
factors (labeled as Modifiers) to aid and simplify modeling processes.

In aid to ensuring high quality input data, values which were incomplete such as timestamps and sample identifiers were searched for and subsequently deleted. Furthermore, rows marked "Missing data" in tenders' Modifier fields were removed as those data represent periods where no valid behavior classification was available.

normal distribution with a standard deviation of 0 (mean) and a standard deviation of 1. This removes the probability of a single feature dominating the model because of scale. Columns that did not add value to the analysis such as samp, ID, and timestamp were removed leaving the dataset clean and behavior labeled.

To understand the class imbalance in the dataset, bar graphs (on a logarithmic scale) were created to examine the distribution of observations across behavioral categories.

Figure 2. Class Distribution of Modifiers (Log Scale)



This step revealed that some behaviors (e.g., rumination) were severely overrepresented compared to rare behaviors such as digging or running.

classification algorithms were implemented: k-Nearest Neighbors (kNN) and Linear Discriminant Analysis (LDA). Both models were trained using the caret package, with hyperparameter tuning and evaluation conducted using k-fold cross-validation.

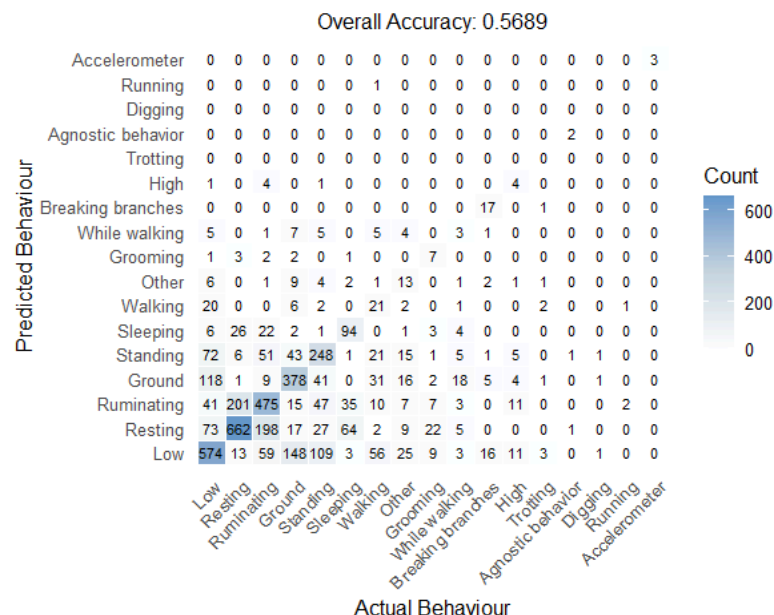
Classification Models

To predict animal behaviors from accelerometer data, two supervised

k-Nearest Neighbors (kNN)

For the kNN model, the dataset was split into training (70%) and testing (30%) sets

Figure 3: Confusion Matrix Heatmap for KNN



using stratified sampling to preserve class distribution. A 5-fold cross-validation was performed on the training set to select the optimal number of neighbors (k). The tuning process iteratively tested values of k from 1 to 10, and the best value was determined based on classification accuracy. Once the model was trained using the optimal k (in this case, k=9), predictions were generated on the test set.

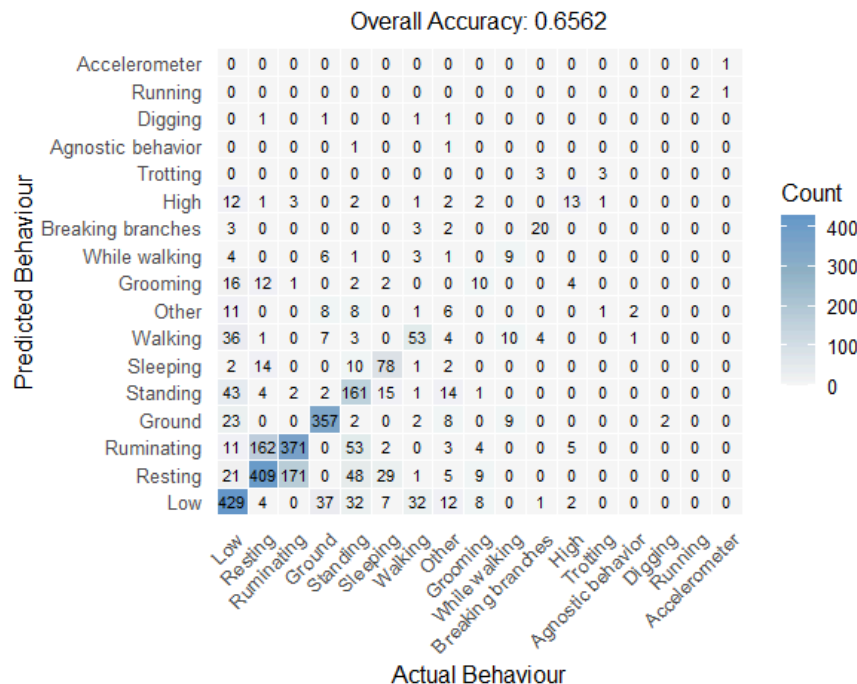
A confusion matrix was computed to evaluate performance, providing insights into both overall accuracy and per-class prediction capability. The final model performance was visualized with a heatmap of the confusion matrix and a plot of classification accuracy versus different values of k. This approach ensures that the

kNN model is not overfitting to the training set, as the repeated cross-validation provides a robust estimate of generalization error.

· Linear Discriminant Analysis (LDA)

To complement the non-parametric nature of kNN, a parametric model, Linear Discriminant Analysis, was also implemented. LDA is particularly suited for problems involving multicollinear predictors, which is likely with accelerometer data that includes derived statistics from correlated axes. Before training the LDA model, an additional feature selection step was performed to improve efficiency and model stability:

Figure 4: Confusion Matrix Heatmap for LDA



1. Near-zero variance predictors were removed using the `nearZeroVar` function from `caret`.

2. Highly correlated predictors (Pearson's $r > 0.95$) were identified and dropped using the `findCorrelation` function to mitigate multicollinearity.

The filtered feature set was then used to train the LDA model. Like with kNN, an 80/20 train-test split was applied, with 5-fold cross-validation on the training set to evaluate model performance. Predictions were made on the test set, and a confusion matrix was computed and visualized using a gradient heatmap. LDA offered an interpretable model with strong performance, particularly valuable in understanding which features best discriminate between different behaviors.

Model Evaluation and Visualization

For kNN and LDA, the performance of classification was assessed by:

- Total accuracy
- Confusion matrices
- Prediction counts per-class

Graphical confusion matrix plots and model tuning

Having these various models provides both flexible (kNN) and interpretable (LDA) perspectives of behavior classification from accelerometer data, and increased confidence in the robustness and reliability of the findings.

3. Results and Discussion

The performance of two model on the test set, can be summarized as below:

Model Test Accuracy

LDA: 65 %

KNN: 56 %

When we check each model's confusion matrix, we see that LDA labels common behaviours (like grazing and walking) correctly more often than kNN, but both models tend to mix up rare behaviours (like digging and running).

The LDA model performed better. We can conclude that the dataset was structured so that simple, linear separators capture most of the behaviour differences, which explains why LDA does better than kNN.

On the other hand, kNN looks at the “nearest neighbours” in all feature directions and then chooses the one with the majority. With so many features and especially when some behaviours have very few examples those chosen neighbours become messy and lead to more mistakes.

Also good to note that we had thousands of clips of some behaviours (like ruminating) but only a few for others (like digging). When a model sees only a small amount of examples for a behaviour, it can’t learn it well and often gets it wrong, pulling down the overall score.

4. Conclusion

Our result shows that an accelerometer gives enough information to predict behaviour well. LDA reached 65 % accuracy and kNN 56 %.

LDA’s better result tells us that, after simple cleaning, the different behaviours fall into groups that straight lines can separate. So we can trust accelerometer data for automatic behaviour monitoring.